



First Semester 2025-2026

M.E. (I Year) Course Handout

Date: 01-08-2026

In addition to Part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course Number : EEE G516
Course Title : Signal Theory and Optimization
Course Instructor : Sainath Bitragunta
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1. Course description:

Probability theory and random variables; joint, conditional, and marginal distributions; functions of random variables; moments, expectation, and statistical inequalities; introduction to random processes; stationarity, autocorrelation, ergodicity, and power spectral density; basic concepts of linear algebra: vector spaces, inner product spaces, linear operators and transformations, spectral decomposition; matrix factorizations including SVD, LU, and QR; positive definite matrices; orthogonal projections in Hilbert spaces; introduction to optimization: convex sets and convex functions; unconstrained and constrained optimization; Lagrange multipliers, Karush-Kuhn-Tucker (KKT) conditions, duality, and numerical optimization techniques.

2. Scope and Objectives:

The instructor presents the content of the course "Signal Theory and Optimization" in five modules. The first module (module-0) comprises preliminaries covering signals classification, definitions, properties, limitations, and applications. The subsequent module (module-1) consists of axiomatic probability theory and random variables, joint, conditional, and marginal distributions of random variables, first and second order statistics, Moment generating function, characteristic function, functions of random variables, and statistical inequalities, generation and simulation of random variables.

The third module (module-2) mainly introduces stochastic processes, specifications, concepts such as stationarity, autocorrelation, crosscorrelation, ergodicity, energy, and power spectral densities. The fourth module (module-3) consists of basic concepts of linear algebra, namely, vector spaces, linear independence and dependence, inner product spaces, linear operators and transformations (matrices), standard matrices (for example, positive definite and semidefinite matrices), matrix decompositions (for example, LU, and QR, Eigen





value decomposition (EVD), singular value decomposition (SVD)) and their uses, Hilbert spaces and norms, projections, inequalities, orthogonalization. The fifth and final module (module-4) consists of an introduction to optimization: convex sets and convex functions; unconstrained and constrained optimization, Karush-Kuhn-Tucker (KKT) conditions, duality, and optimization techniques such as Lagrange multipliers, numerical optimization (non-convex) techniques.

The objectives of the foundational course are to lay strong foundations for i) probability and random signals and related theory, ii) linear algebra, matrices iii) optimization techniques and their uses in communications and signal processing.

3. Course Outcomes:

At the end of the course Signal theory and optimization, students learn useful mathematical and probability concepts, tools (transforms, matrices, optimization), and techniques to model several phenomena that occur in communication systems and networks, and analyze and design them better. Specifically, they learn the significance of signal theory, fundamental laws of probability, finding moments of random variables, modeling using random variables, and random processes and their applications. Furthermore, generation and simulation of random variables using software tools like MATLAB/Python are also one of the essential practical outcomes of this course. In addition to strong random signal theory foundations, students learn useful techniques of linear algebra and matrices, optimization, which are useful in understanding and analyzing machine learning-aided next generation communication networks.

4. Text Books

[T1]. A. Papoulis and S. U. Pillai, Probability, Random Variables and Stochastic Processes, 4th ed., McGraw-Hill Education, 2002.

[T2]. G. Strang, Linear Algebra and Its Applications, 4th ed., Cengage Learning, 2007.

[T3]. Boyd and L. Vandenberghe, Convex Optimization, Cambridge University Press, 2004.

Reference Books

[R1] Introduction to probability, D. P. Bertsekas, J. N. Tsitsiklis, second edition, Athena scientific, 2008.

[R2] Discrete stochastic processes by R. G. Gallager, Springer, 1996.

[R3] Random processes for engineers, Bruce Hajek, Illinois, 2014.





[R4] Intuitive probability and random processes using MATLAB, S. M. Kay, Springer, 2006.

[R5] Todd K. Moon and Wynn C. Stirling, Mathematical Methods and Algorithms for Signal Processing, Prentice Hall, 1999.

5. Method of conduct of the course: offline mode, lectures include mostly blackboard work and partly PowerPoint presentations.

6. Course Plan

Lect. No.	Topics to be covered	Learning Objectives	Source
Module 0 (1-3)	Preliminaries: classification, definitions, limitations and applications of various signals	Students could identify, classify, understand and analyze different signals (for example, deterministic, pseudorandom and stochastic), get to know their applications in communications and signal processing	[T1], [R5], [R3], Tutorial papers (e.g., IEEE)
Module 1 (4-6)	Axiomatic probability, various concepts and simple rules/theorems	Students understand and learn probability space for simple random experiments, application of axioms to prove lemmas, theorems, pictorial representations, motivating examples	[T1]/[R1]/[R3]
Module 1 (7-12)	Random variables: continuous (CRV), discrete (DRV), and mixed (MRV), cumulative distribution function (CDF), probability density function (PDF), probability mass function (PMF), Moment generating function (MGF) &	Students could differentiate between continuous and discrete random variables, determine first order and second order statistics of random variables, significance, application of MGF, application of inequalities, applications	[T1]/[R3]





	Characteristic function, properties, Useful inequalities, Central limit theorem, applications, examples		
Module 1 (13-21)	Random variables (Contd.): transformation of random variables, generation and simulation of random variables, simulation of random variables, multiple random variables	Affine transformation of continuous random variables (RVs), non-linear transformation of RVs, generation of Gaussian, Rayleigh RVs from uniform RV, uncorrelated and correlated RVs, statistical independence of RVs, orthogonality, simulation of some RVs, Cauchy-Schwarz inequality, bivariate and multivariate Gaussian distribution,	[T1]/[R3]/[R4]
Module 2 (22-27)	Stochastic processes: Specifying random process, types of random process, wide sense stationarity (WSS), ergodicity, transmission of random process, examples, applications	Specifying random process, examples, WSS, stationarity in strict sense, conditions for ergodicity, Energy and power spectral densities, Gaussian random processes and their properties.	[T2]/[R2]/[R3]
Module 3 (28-30)	Linear algebra and matrices	Students learn concepts on vector spaces, linear independence and dependence, inner product spaces, inequalities	[T2], [R5]
Module 3 (29-33)	Linear algebra and matrices	Students could understand linear operators and transformations (matrices), standard matrices (for example, positive definite and semidefinite matrices), and apply matrix decompositions (for	[T2], [R5]





		example, LU, and QR, EVD, SVD), Hilbert spaces and norms, projections, inequalities, orthogonalization	
Module 4 (34-38)	Foundations for optimization (concepts, tools)	Students could understand convex sets and convex functions; unconstrained and constrained optimization, Karush-Kuhn-Tucker (KKT) conditions, duality	[T3]/[R5]
Module 4 (39-43)	Convex and non-convex optimization: key techniques, limitations and few applications	Lagrange multipliers approach, numerical optimization (non-convex) techniques, applications	[T3]/[R5]/Online resources

7. Laboratory component: Lab exercises complement the understanding of various topics in probability, random variables and random processes, linear algebra and Matrix theory, optimization techniques. These exercises will mainly involve simulations using MATLAB/Python programming. Additional details will be announced in the class/posted on Nalanda (LMS portal).

8. Evaluation Scheme*

Component	Duration	Weightage	Date & Time	Venue	Remarks
Quiz	40 minutes	10%	To be announced (TBA)	Classroom	Closed book (CB)
Mid-Semester Test	90 minutes	20%	Oct. 6 th (FN2)	TBA	CB
Labs & Class tasks	TBA	10% + 5%	TBA		Open book (OB)
Team project*	TBA	15%	TBA		OB
Comprehensive Examination	3 hrs.	40%	Dec. 5 th (AN)	TBA	OB/OB+CB

*Seminar: 5% ; Work done and Report: 10%





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Instruction Division

9. Chamber Consultation Hour and announcements: To be announced in the class/LMS (Nalanda).

10. Notices: All the notices will be posted on Nalanda (LMS portal).

11. Makeup Policy:

No makeup exam will be conducted for quizzes. However, for the mid-semester and the comprehensive exam, makeup will be conducted for valid reasons with supporting documents and prior permission.

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